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Senior Industrial Control Systems & Automation Specialist Project Portfolio

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⇒ **DTA File Hub · Full-Stack Secure Client File Delivery Portal**

April 2026 – Present [#45](#)

☐ **URL:** files.dt-automation.net

➤ **Full-Stack Web Application · Cloud Infrastructure · DevOps**

- Conceptualized, independently architected, developed, tested and deployed DTA File Hub — a full-stack secure file management and client delivery portal at files.dt-automation.net — built on Cloudflare infrastructure (Workers, Pages, R2, Zero Trust Access) with Resend email delivery and Svix webhook verification. Features a multi-role system (Admin, Accounting, Client), 27 API endpoints, serial-numbered client delivery workflow (DTA-DEL-CLTxxx-YYYY-NNN), persistent activity logging with 11 event types, real-time email status badges, PIN-protected client delivery page with ZIP download, dynamic space access management with no redeploy required, and subfolder navigation. Running cost: €0/month on free tiers. Total: ~28 hours of design, development, testing and deployment.
- **Tools & Technologies:** Cloudflare Workers (JS) · Cloudflare R2 · Cloudflare Pages · Cloudflare Zero Trust · Resend API · Svix · JSZip · Claude.ai (Anthropic)

⇒ **DTAutomation · Bilingual Professional Portfolio & Brand Platform**

March 2026 – Present [#44](#)

☐ **URL:** dt-automation.net · dt-automation.gr

➤ **Web Platform · Brand Development · Front-End Engineering**

- Conceptualized, designed, developed, tested and deployed a professional bilingual (EN/GR) portfolio & brand platform at dt-automation.net — encompassing full Brand Name Development and independently architected on Carrd.co Pro Plus with a custom 15 Body End embed JS architecture, inter-embed global bridge system, slide-in project modal with structured data rendering, GLightbox + Plyr video system with muted autoplay deeplink support, 43-project portfolio with persistent row highlight, LinkedIn-compatible video & project deep link infrastructure via GitHub Pages (12 video deep links + 43 project deep links with OG tags and project-specific thumbnails), Web3Forms + hCaptcha contact system, full SEO / Open Graph / Schema JSON-LD implementation across both domains, and site protection. Total: ~32 hours of design, development, testing and deployment.
- **Tools & Technologies:** Carrd.co Pro Plus · HTML5 · CSS3 · JavaScript (ES5/Vanilla) · GitHub Pages · GLightbox · Plyr.js · Python (ReportLab, Pillow) · Web3Forms · hCaptcha · JSON-LD / Schema.org · Claude.ai (Anthropic)

⇒ **Applied R&D, Full-Stack, Life Cycle, Continuous Improvement Development: Machinery Operation Philosophy Enhancement & Design, Integration, Configuration, Programming, Commissioning & Functional Integrity Verification (FAT/SAT) of all underline Industrial Automation Equipment involved (PLC, HMI, Industrial Robots, Servo Drives, Stepper Drives, VFDs).**

Jan 2020 – Apr 2026 [#43](#)

❑ Integration Of:

- ❑ MITSUBISHI MELSEC iQR PLC (04, 08, 16)
- ❑ MITSUBISHI MELSEC iQR I/O, High Speed I/O, FPGA-Based Flexible High Speed I/O Control
- ❑ MITSUBISHI MELSEC iQR Serial RS-485 Communication Modules (RJ71C24-R4)
- ❑ MITSUBISHI MELSEC iQR Simple Motion Module (RD77MS16)
- ❑ MITSUBISHI MELSERVO (MR-J4, MR-J5)
- ❑ MITSUBISHI VFDs (FR-A800-E in Position Vector Control, FR-D700, FR-E800)
- ❑ MITSUBISHI HMI (GOT2000)
- ❑ MITSUBISHI MELFA Industrial Robot (CR800-CVD, RV-8CRL-D-S15)
- ❑ MITSUBISHI MELSEC iQF PLC (FX5UC)
- ❑ MITSUBISHI MELSEC iQF I/O, High Speed Counter
- ❑ LIKA Integrated Closed Loop Stepper Controllers RD5
- ❑ IAI Servo Controller PCON

- Successfully developed, accomplished and delivered: ⇒ **4 new machines** ⇒ **One of which includes the combined integration of a 6-Axis articulated industrial Robot** ⇒ **Continuous iterations/improvements of these machines.**
- Successfully delivered and commissioned: ⇒ **18 machines both locally and abroad.**
- Successfully only-developed: ⇒ **2 new machines.**
- Successfully conceptualized, generated, designed, developed, implemented, standardized and delivered **Advance Synchronous Multi-Axis Motion Control** algorithms and methodologies, using real time configurable, software-based and User Operator friendly digital **CAM control**, including **Real-Time Automatically Adjustable** and production rate based **Phase Compensation** algorithms, in order to achieve higher product quality, precision, increased production rates, effortless machine configuration & product setup, reduced machine: acoustic noise and mechanical wear & tear.
- Successfully conceptualized, generated, designed, developed, implemented and delivered a **Robotics Application** in which, **Abstractive Projected Trigonometric Methodologies**, along with “**Frame**” manipulation techniques, have been implemented as software in the Robot’s Controller. Through this implementation the End User Operator can effectively, efficiently and quickly, create through the machine’s HMI, recipes for variable sized Single Crate Pair Palletizing, by entering only a few crate dimensions. With this **solution approach**, the customer’s need for qualified and Robot Programming & Handling trained personnel has been eliminated.

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⇒ Single Servo-Axis Flow Pack Machine.

Jun 2019 [#42](#)

Project description:

- ☐ Integration of:
 - ☐ SIMATIC S7-1200 PLC.
 - ☐ 1 PLC Communication Module for CANOpen.
 - ☐ 1 Servo-Controller (LTI) (CANOpen).
 - ☐ 1 MITSUBISHI VFD.
 - ☐ 1 SIMATIC Basic HMI.
- Project included:
 - Adaptation of compile-time PLC code, to accordingly match the machine's mechanical data.
 - HMI programming for additional operational features.
 - Commissioning.

⇒ Automated Metal Hinge Drilling and Tapping Processing Machine.

Feb 2019 – Apr 2019 [#41](#)

Project description:

- ☐ Integration of:
 - ☐ 1 SIMATIC S7-300 PLC.
 - ☐ 3 MITSUBISHI VFD.
 - ☐ 1 Servo-Controller (LTI) (Pseudo Servo-Control for 3Phase AC Induction Motor) (Profibus)
 - ☐ 1 GMT HMI Panel
- Process Automation System Development, Integration, On-Site Wiring, Configuration Implementation, Programming, Commissioning and Functional Testing.

⇒ Automated Sheet Metal Feeder & Iron Casting Press Machine.

Sep 2018 – Nov 2018 [#40](#)

Project description:

- ☐ Integration of:
 - ☐ Mitsubishi iQ-F PLC FX5 CPU.
 - ☐ 1 MITSUBISHI FR-A800-E VFD.
 - ☐ 1 MITSUBISHI GOT HMI Panel.
- Mitsubishi FR-A800-E Inverter configured to work in "Position Control Mode" under "Vector Control", with encoder interface option unit installed (Pseudo Servo-Control for 3Phase AC Induction Motor).
- CC-Link IE Field Basic Network interface between FX5 CPU and FR-A800-E Inverter.
- Process Automation System Development, Integration, On-Site Wiring, Configuration Implementation, Programming, Commissioning and Functional Testing.

⇒ A Proprietary Application User Interface Tool For Calibrating And Evaluating The Taught Points Inside A Fanuc Teach Pendant Program.

Jul 2018 – Aug 2018 [#39](#)

Project description:

- Problem Case: Mechanical inconsistencies were observed at the End Of Arm Tool (TIP-TIG Torch), while the tungsten consumable was changed by the operator, either for maintenance or grinding purposes. After the replacement of the tungsten, the result was a “different” actual Tool Center Point, due to the misplacement of the tungsten while it was being tighten. With limited available resources, the only way to compensate this misplacement was to reteach the robot’s process points or to develop a flexible, yet easy, feasible and fast, for the inexpert end-user operator procedure, to recalibrate them.
- Project included:
 - Design, development, programming and implementation of an Application User Interface Tool to aid both the integrator and the end-user operator to recalibrate the process’s taught points.
 - Development, Programming and Simulation was implemented in Fanuc ROBOGUIDE IDE, using KAREL programming language.
 - Programming included:
 - Teach Pendant Program file manipulation, to individually and selectively read the taught points from different .TP files of the process.
 - Functions to selectively move the robot, at the desired taught point and adjust it by entering the calibrated offset values by hand.
 - Functions to run and evaluate the predetermined motion segments of the process, which include both the calibrated and original taught points.

⇒ Automated Robotic Cell For Weight Plates Distribution

Jun 2018 – Aug 2018 [#38](#) 

Project description:

- ☐ Integration of:
 - ☐ SIMATIC S7-1200 PLC.
 - ☐ GMT HMI Panel.
 - ☐ Fanuc 6-Axis Industrial Robotic Arm (Robot Model: S420L-iF) (Profibus).
- Process Automation System Development, Integration, On-Site Wiring, Configuration Implementation, Programming, Commissioning and Functional Testing.
- Robot Process Development, Integration, Implementation, Programming, Commissioning and Functional Testing.
- Fanuc Robotics (Robot Model: S420L-iF): Machine Tending, Pick And Place Application Process.

⇒ Troubleshooting & Maintenance Of An Automated Silicone Tube Production Machine.

Apr 2018 [#37](#)

Project description:

- Project included:
 - Preliminary diagnosis of the defective process (Depleted PLC battery caused the deletion of program and data, from the memory of the PLC).
 - Preliminary work upgrading the obsolete PLC project source files, from MITSUBISHI MELSEC MEDOC to MITSUBISHI GxWorks 2 format.
 - Re-Commissioning of the system.

⇒ Operational Functioning Review Of A Theater Automation System Infrastructure And Overall Optimization Of Its Underlying PLC And HMI Software In Order To Comply Accordingly With The Standards Of Din 56950-1 And IEC 61508 (SIL3).

Sep 2017 – Dec 2017 [#36](#)

Project description:

- Revision, Correction, and ultimately Optimization of both hardware configuration and the accompanying software of all of the underlying devices of the system.
- The system under investigation is mainly comprised of:
 - 2 SIMATIC Safety Integrated S7-300-F CPU's.
 - 20 MITSUBISHI VFD's, with encoder interface optional cards, for closed loop position control (Profibus).
 - 1 SIMATIC WinCCFlexible HMI Panel.
- Conclusively, the investigation of the system resulted in the complete re-design, re-programming and re-implementation of the software of the PLC's and HMI.
- The overall effort of this undertake, and after the official inspection by the TUV, resulted in a system which complies with SIL3 Functional Safety, by 95%. By contrast, the system was previously in such a condition, in which the inspection tests from the TUV, could not be even performed.
- Project included, the written Risk Analysis of all of the Safety Functions that were implemented by the Control and Monitoring system.
- Process Automation System Development, Integration, On-Site Wiring, Configuration Implementation, Programming, Commissioning and Functional Testing.

⇒ 3 Servo-Axis Flow Pack Machine.

Dec 2017 [#35](#)

Project description:

- ☐ Integration of:
 - ☐ 1 MITSUBISHI iQ-F PLC FX5 CPU.
 - ☐ 3 MITSUBISHI RJ-4 Servo Drives.
 - ☐ 1 MITSUBISHI GOT HMI Panel.
- Project Included:
 - On-Site I/O Crosscheck and validation of electrical wiring.
 - HMI programming for additional operational features.
 - Commissioning.
 - Customer machine operation training.

⇒ Re-Design, Re-Program And Re-Implementation Of A Process Automation System For The Transportation And Supply Of Empty Cans To A Canning Plant's Fruit Filling Stations.

Jun 2017 - Jul 2017 [#34](#)

Project description:

- Project included the re-programming of the PLC software code, aimed at improving and optimizing the overall underlying process, both for the current infrastructure as well as for future updates and additions.
- Consequently, the PLC software was rebuilt based on configurable at compile-time, generalized and abstracted Function-Blocks, so as if an addition of a new supply line is needed, the down time to be a matter of just a day's time span.
- ☐ Underlying hardware included:
 - ☐ 1 S7 – 300 CPU

⇒ Obsolete Fruit Juice-Puree Aseptic/Filling Process Automation Machine Maintenance, Troubleshooting & Commissioning.

Jun 2017 [#33](#)

Project description:

- Task included:
 - The complete maintenance and adaptation of the Underlying control system of an obsolete Aseptic Machine, targeting effective and appropriate overall process operation.
 - Reverse Engineering of the actual electrical wiring circuits, in order to actually implement the required modifications and corrections.
 - Reverse Engineering of OMRON PLC code (without having the original engineering project at our disposal), so as to proceed and implement the appropriate modifications and corrections of the process automation functional characteristics.

⇒ Asphalt Process Automation Production Plant Troubleshooting.

Feb 2017 - Mar 2017 [#32](#)

Project description:

- Project included the troubleshooting of the underlying SCADA software (Wonderware InTouch), due to the fact that process data, could not be archived in the local database, for inexplicable reason.
- After resolving the issue in the SCADA software, we realized that there was hardware malfunction, in the wireless process data transmission modules as well, due to improper configuration.
- Reverse Engineering both in hardware as well as in software level.

⇒ Automated Robotic Cell For MIG Welding Process.

2016 – 2017 [#31](#) 

Project description:

- ☐ Integration of:
 - ☐ SIMATIC S7-1200 PLC.
 - ☐ GMT HMI Panel.
 - ☐ Fanuc 6-Axis Industrial Robotic Arm (Robot Model: S420-iF) (Profibus).
 - ☐ Microprocessor-controlled Multi-Process Welding Machine.
- Process Automation System Development, Integration, On-Site Wiring, Implementation, Programming, Commissioning and Functional Testing.
- Robot Process Development, Integration, Implementation, Programming, Configuration Commissioning and Functional Testing.
- System designed for various and different types of products, with expandable and dynamic features regarding the addition of new casts-products.
- Fanuc Robotics (Robot Model: S420-iF): Automated PULSED MIG Welding Process (Steel).

⇒ Design, Development And Implementation Of A Real-Time And Fault-Tolerant ModBus RTU Master Function-Block Library For Mitsubishi iQ-F FX5 PLC.

2017

⇒ Troubleshooting, Maintenance & Optimization Of A Fruit Juice/Puree Aseptic/Filling Process Automation Machine.

Nov 2016 [#29](#)

Project description:

- Project included:
 - Troubleshooting: Problem case was a bug in the PLC software code that had existed in the automated process since its first commissioning. While the system state was being in normal production phase, and just before filling the product in the barrel bags, if the sensor in the cap remover sub system, could not detect the cap, for whatever mechanical or electrical reason, then the process was being ended up into a deadlock.
 - Result of the problem was significant production downtime, in the scale of one shift change, about 8 hours. Process was able to restart again, after the CIP sub-system had completed its full cleaning cycle and then again after the asepsis procedure had finished.
 - Off-Site preliminary work included, the update of the source files of the PLC software code to add information about the I/O signals of the process (original project source files were not at our disposal), as well as some preparatory reverse engineering.
 - On-Site reverse engineering and modification had been performed, in the PLC software code. (2 – 3 hours).
 - Ultimately, the problem not only had been solved without any drawbacks, but with the production downtime actually being eliminated to zero.
- ☐ Underlying hardware comprised of:
 - ☐ 1 SIMATIC S7-300 PLC.

⇒ Maintenance And Update Of A Fruit Juice/Puree Aseptic/Filling Process Automation Machine.

Nov 2016 [#28](#)

Project description:

- Problem consisted of system's current operator panel functional defect in hardware level. (Uniop Operator Panel)
- Project included:
 - Reverse engineering in the software code of Uniop Operator Panel.
 - The re-installation, re-programming and re-commissioning of the system's original HMI panel (SIMATIC HMI panel), that was previously replaced by the Uniop Operator Panel.
 - Correction of TAGS, correction and reconfiguration of Alarms, visual interface corrections and modifications (SIMATIC ProTool/Pro CS).
- ☐ Underlying hardware comprised of:
 - ☐ 1 SIMATIC S7-300 PLC.
 - ☐ 1 SIMATIC HMI Panel.

⇒ Motion Control System Upgrade Of "Pack Line PXM Filling And Sealing Machine".

Oct 2016 [#27](#)

Project description:

- Project included:
 - The replacement of an obsolete Control Techniques Unidrive with a DELTA ASDA-A2 Servo Drive.
 - On-Site electrical wiring of the digital control signals of the Servo Drive.
 - Configuration of the Servo Drive and adaptation of the digital control signals in accordance with the old wiring.
 - Commissioning and functional testing of the Servo Drive.

⇒ Automated Robotic Cell For "Assembly & Welding Of Vertical Type Micro-Wind Turbine Parts". 2014 – 2016 [#26](#)

Project description:

- ☐ Integration of:
 - ☐ 1 SIMATIC S7-300 PLC.
 - ☐ 1 SIMATIC HMI Panel.
 - ☐ 6 Servo-Controllers (LTI) (Profibus).
 - ☐ 2 Distributed Valve Terminals (Festo/CPX) (Profibus).
 - ☐ 2 VFD SIEMENS MICROMASTER.
 - ☐ 1 Fanuc 6-Axis Industrial Robotic Arm (Robot Model: S420-iF) (Profibus).
 - ☐ 1 Microprocessor-controlled TIP-TIG Hot Wire Welding Machine.
- Process Automation System Development, Integration, On-Site Wiring, Implementation, Programming, Configuration, Commissioning and Functional Testing.
- Robot Process Development, Integration, Implementation, Programming, Commissioning and Functional Testing.
- Including the development and implementation of a proprietary Fault-Tolerant Function-Block library for driving the Servo-Controllers, through Profibus.
- Fanuc Robotics (Robot Model: S420-iF): Automated TIP-TIG Hot Wire Welding Process (Stainless Steel, Steel).

⇒ Design, Development And Implementation Of Function-Block Library For Driving Delta Inverters Through Modbus With Simatic S7-1200 PLC. 2016

⇒ Design, Development And Implementation Of Function-Block Library For Driving Solocom & Hapn VFD's Through Modbus With Simatic S7-1200 PLC. 2016

⇒ Improvement And Optimization Of Animal Food Production Process. 2016 [#23](#)

Project description:

- Project included, the improvement and optimization of the recipe batch dosage sub-system of the production plant. The overall undertake comprised of non-invasive modification and adaptation of the PLC and SCADA software, in order to combine the main dosage recipe and the micro – dosage recipe to a complete and unified recipe, resulting in the continuous and flawless operation of the overall production process, without interruption.
- Reverse Engineering of the PLC and SCADA software.
- Additional SCADA software was added, for visualization, monitoring and automated reporting of the data of the combined recipe batches.
- From the resources at hand, the original project of the PLC code was not at our disposal.
- ☐ Underlying hardware and SCADA included:
 - ☐ 1 S7 – 300 CPU
 - ☐ 3 Distributed ET 200M Stations
 - ☐ WinCC Standalone V7.0

⇒ Industrial Automation Infrastructure Upgrade At The Facilities Of Biological Wastewater Treatment Of Thessaloniki (Greece).

2015 – 2016 [#22](#)

Project description:

- ❑ Integration of:
 - ❑ Mitsubishi iQ platform Programmable Logic Controllers.
 - ❑ VFDs: Mitsubishi, ABB
 - ❑ Various Instrumentation and Metering devices
- Process Automation System Implementation, I/O Cross-Check, Programming, Commissioning and Functional Testing for the following stages:
 - Wastewater Inlet Pump Station.
 - Wastewater Screening Process.
 - Dehydration Process.
- ❑ Integration of Mitsubishi L platform PLC and multiple VFDs in PID Loop Configuration.
- Process Automation System Implementation, Configuration, Programming, Commissioning and Functional Testing for the following system:
 - Degradation of the underline aquatic horizon, for safe maintenance operations at the waste water's sedimentation reservoirs.

⇒ Relocation Of Customer's Industrial Premises. Re-Installation And Re-Commissioning Of Robotic Cells.

(Work Abroad) Jun 2015 [#21](#)

Project description:

- ❑ Underlying hardware included 2 Industrial Robotic Cells constituted of:
 - ❑ 1 SIMATIC S7-1200 PLC.
 - ❑ 1 SIMATIC HMI Panel.
 - ❑ 1 Fanuc 6-Axis Industrial Robotic Arm (Robot Model: S420-iF) (Profibus).
- Project Included:
 - Coordination and execution of the re-installation and On-Site re-wiring of the underlying hardware.
 - Robot Process re-programming (upgrade/optimization).
 - Re-commissioning.

⇒ Design, Development And Implementation Of A Real-Time And Fault-Tolerant Modbus RTU Master Function-Block Library For Simatic S7-1200 PLC.

2015

⇒ **WinCC V7.2 (Standalone) Configuration And Commissioning.**
(Facilities Of Biological Wastewater Treatment Of Thessaloniki, Greece)
2015 #19

Project description:

- Project included:
 - Configuration and Commissioning of 1 Redundant Pair of WinCC Servers.
 - Configuration and Commissioning of 2 WinCC Clients for the Redundant Pair of WinCC Servers.
 - Installation, Configuration and Commissioning of:
 - 1 WinCC/WebNavigator Server in one of the WinCC Clients (PC 1).
 - 2 WinCC/WebNavigator Clients in each one of WinCC Clients.
 - 1 WinCC/DataMonitor Server in one of the WinCC Clients (PC 1).
 - 1 WinCC/DataMonitor Client in one of the WinCC Clients (PC 1).
 - Configuration and Commissioning of the WinCC/DataMonitor Client, for producing Reports.
 - Configuration of the WinCC Server User Administration.
 - Configuration of the underlying windows operating systems of all the PCs that got involved namely:
 - 2 x Windows Server 2008 R2 Standard SP1 64-bit (Servers)
 - 2 x Windows 7 SP1 32-bit (Clients).
- ☐ The above architectural structure and combination of topology was proposed from the author of this document.
- Configuration and Commissioning of WinCC OPC Server.
- Configuration and Commissioning of SIMATIC PC Station and S7Connections (OPC Communication for S7-1200) for the underlying hardware, using SIMATIC NetPro.
- Provided Direct Training to the Operator and Maintenance Officer Engineer of the SCADA, for the above aforementioned configuration and commissioning procedures.

⇒ **Cartesian X – Y Coordinated MIG Welding Machine.**
Mar 2014 – Apr 2014 #18

Project description:

- ☐ Integration of:
 - ☐ SIMATIC S7-1200 PLC.
 - ☐ SIMATIC HMI Panel.
 - ☐ 2 "Closed Loop" Stepper Controllers (FESTO) (Profibus).
- Process Automation System Development, Integration, On-Site Wiring, Implementation, Configuration, Programming, Commissioning and Functional Testing.

⇒ **Automated Rod Metal Feeder Machine.**
Jan 2014 – Mar 2014 #17

Project description:

- ☐ Integration of:
 - ☐ SIMATIC S7-1200 PLC.
 - ☐ SIMATIC HMI Panel.
 - ☐ 1 Servo-Controller (LTI) (Pseudo Servo-Control for 3Phase AC Induction Motor) (Serial Communication Interface).
- Process Automation System Development, Integration, On-Site Wiring, Implementation, Configuration, Programming, Commissioning and Functional Testing.

⇒ Automated Sheet Metal Feeder-Extrusion Machine.

Jan 2014 – Mar 2014 [#16](#)

Project description:

- ☐ Integration of:
 - ☐ SIMATIC S7-1200 PLC.
 - ☐ SIMATIC HMI Panel.
 - ☐ Servo-Controller (LTI) (Pseudo Servo-Control for 3Phase AC Induction Motor) (Serial Communication Interface).
- "Full Close Loop" Servo-Controller Operation with a secondary encoder placed at the final output after the extrusion process.
- Process Automation System Development, Integration, On-Site Wiring, Implementation, Configuration, Programming, Commissioning and Functional Testing.
- Including the development and implementation of a proprietary Real-Time and Fault-Tolerant serial communication interface protocol Function-Block library for driving the Servo-Controller.

⇒ Automated Robotic Cell For Material Handling Of "Sliding Gate Metal Wheels" Cooperative CNC Production Process.

Oct 2013 – Jan 2014 [#15](#) 

Project description:

- ☐ Integration of:
 - ☐ SIMATIC S7-1200 PLC.
 - ☐ SIMATIC HMI Panel.
 - ☐ Fanuc 6-Axis Industrial Robotic Arm (Robot Model: S420-iF) (Profibus).
- Process Automation System Development, Integration, On-Site Wiring, Implementation, Configuration, Programming, Commissioning and Functional Testing.
- Robot Process Development, Integration, Implementation, Programming, Commissioning and Functional Testing.
- Project included, the replacement of an obsolete Cartesian Robotic System with the aforementioned articulated industrial robotic arm and generally, the upgrade of the overall Industrial Automation Infrastructure of the manufacturing process, along with process reverse engineering.
- PLC & Robot Process Software Code designed and implemented in such a way so as, not any intervention is required from the inexperienced end-user operator, in order to change the part being processed by the system. Robot positional offsets are used in the overall PLC & Robot Process Software Code, so as to enable the operator to change process parts from the HMI of the system.
- Fanuc Robotics (Robot Model: S420-iF):
 - Machine Tending Application Process
 - Pick And Place Application Process

⇒ A Custom Automated Industrial Machine For "Functional & Material Endurance Test Of Castors".

Aug 2013 – Oct 2013 [#14](#)

Project description:

- ☐ Integration of:
 - ☐ SIMATIC S7-1200 PLC.
 - ☐ SIMATIC HMI Panel.
 - ☐ 2 Delta Inverters, controlled through Modbus RTU protocol communication interface.
- Process Automation System Development, Integration, On-Site Wiring, Implementation, Configuration, Programming, Commissioning and Functional Testing.

⇒ Total SCADA Upgrade Of Industrial Process Automation Plant For Animal Food Production. 2013 [#13](#)

Project description:

- Upgrade/Migration of obsolete version of Wonderware InTouch SCADA running on a Windows 98 machine, to a newer version of the same SCADA platform on a Windows 7 machine.
- System included 1 Server and 1 Client application, Simatic S5, S7 Programmable Controllers.
- Reverse Engineering in software level.

⇒ “A Discrete Electronics Emulation Model Of Photovoltaic Cell”. Mar 2012 – Aug 2012 [#12](#)

Project description:

- Successful key-role applied research in the development, enhancement and actual implementation of a "Discrete Electronics Model" which emulates the full-scale range of the operational characteristics of a photovoltaic cell.
- Software-based Simulation.
- Analogue Electronics Circuit-Hardware Design, Manufacture and Implementation.

⇒ Design And Implementation For Manufacture Of A Power Supply, Based On Recycled Electronics, For Laboratory Usage. Mar 2012 – Aug 2012

⇒ MSP430 2.4GHz USB Dongle. Sep 2010 – Oct 2010 [#10](#)

Project description:

- ☐ Electronics Product Design, Development & Realization.
- ☐ Hardware Design.
- ☐ Printed Circuit Board (PCB) Design for Manufacture.
- An MSP430 micro-controller based electronic system that incorporates a low power, short range RF transceiver. The system can be directly connected with a PC, through USB interface and can be used for wireless communication.
- The system also features on-board JTAG interface as well as hardware GPIO, for programming, debugging and testing within the development cycle of any custom application. It has been designed in such a way so as, the extra PCB area, featuring the aforementioned evaluation hardware block, to be extracted from the design files, with minimal changes in the actual application hardware blocks.
- The size and shape of the 2-Layer PCB matches the mechanical specifications of a chosen USB Dongle plastic case.
- It can be used for various low power-based wireless applications including wireless sensor networking. The MSP430 device embodies an integrated USB interface, minimizing the cost and allows developers, to make firmware updates on the fly.

⇒ “An Autonomous Low Power Telemetry System For Temperature Monitoring And Data Logging”.

Sep 2008 – Sep 2009 [#09](#)

Project description:

- ☐ Electronics Product Design, Development & Realization.
- ☐ Embedded System Design & Development For Manufacture.
- ☐ Hardware Design, Printed Circuit Board (PCB) Design For Manufacture.
- ☐ Embedded Software Design.
- ☐ TI MSP430 Assembly Software Programming
- The overall system features wireless inter-connectivity among a number of portable data loggers and a monitoring subsystem which comprises, a PC running windows-based data acquisition software and a wireless access point. The main work was focused on the data logger, an electronic sensing and data acquisition module.
- The module incorporates, mixed-signal and ultra-low power micro-controller technology, along with a digitally controlled, low power, short range RF transceiver so as to enable remote system configuration and logged data transmission.
- Two prototype wireless temperature data loggers were manufactured and tested. In order to test the wireless communication channel, as well as to implement it and evaluate overall system's functional performance, a simple supervisory and data acquisition software was developed.

⇒ **MSP430 Micro-controller Based EEPROM Selector.**

2008 [#08](#)

Project description:

- ☐ Electronics Product Design, Development & Realization.
- ☐ Embedded System Design & Development For Manufacture.
- ☐ Hardware Design, Printed Circuit Board (PCB) Design For Manufacture.
- ☐ Embedded Software Design.
- ☐ Embedded C Programming.

⇒ **Domestic Security Alarm System.**

2005 – 2008 [#07](#)

Project description:

- ☐ Electronics Product Design, Development & Realization.
- ☐ Printed Circuit Board (PCB) Design For Manufacture.
- ☐ Based on 8051 Micro-controller.

⇒ **MS Windows Industrial Software Application For Temperature Sensor Data Logging.**

2005 – 2008 [#06](#)

Project description:

- ☐ Electronics Product Design, Development & Realization.
- ☐ Software Development and Implementation in Microsoft Visual Studio (.NET Framework).
- ☐ Serial Communication Interface to an embedded electronic system.
- ☐ Sensor management, Data Logging.

⇒ OneWireBus Coupler/Repeater Node.

2005 – 2008 [#05](#)

Project description:

- ☐ Electronics Product Design, Development & Realization.
- ☐ Hardware Design, Printed Circuit Board (PCB) Design For Manufacture.
- A dual OneWireBus Slave Node for remote relay contact control, with OneWireBus signal repeater capabilities.

⇒ OneWireBus Master/Modbus RTU Controller Bridge.

2005 – 2008 [#04](#)

Project description:

- ☐ Electronics Product Design, Development & Realization.
- ☐ Embedded System Design & Development For Manufacture.
- ☐ Based on MSP430 micro-controller.
- ☐ Hardware Design, Printed Circuit Board (PCB) Design.
- ☐ Embedded Software Design.
- ☐ Embedded C Programming.

⇒ Photovoltaic Array Instrumentation Electronics.

2005 – 2008 [#03](#)

Project description:

- Theoretical Hardware & Printed Circuit Board (PCB) Design For Manufacture.

⇒ RS232 To RS485 Serial Converter.

2005 – 2008 [#02](#)

Project description:

- ☐ Electronics Product Design, Development & Realization.
- ☐ Hardware Design, Printed Circuit Board (PCB) Design For Manufacture.

⇒ Water Level Detection and GSM Controller.

2005 – 2008 [#01](#)

Project description:

- ☐ Electronics Product Design, Development & Realization.
- ☐ Hardware Design, Printed Circuit Board (PCB) Design For Manufacture.
- ☐ Based on 8051 micro-controller.